Project Design Phase-II

Data Flow Diagram & User Stories

Date	**13 September 2026**
Team ID	
Project Name	**Medical Diagnostic Center – ServiceNow**
Maximum Marks	**4 Marks**

Data Flow Diagram (DFD)

A Data Flow Diagram represents how information moves through the **Medical Diagnostic Center** system. It shows the interaction between patients, diagnostic center staff, the ServiceNow application, payment services, and laboratory reports.

Level 0 – Context Diagram

External Entities:

- Patient
- Doctor / Lab Technician
- Administrator
- Payment Gateway
- Email/Notification Service

Main System: Medical Diagnostic Center – ServiceNow

Major Data Flows:

- Patient → Test search, booking request, patient details
- System → Appointment confirmation, appointment status, laboratory reports
- Payment Gateway ↔ Payment details and payment status
- Doctor/Lab Technician ↔ Test results and report verification
- Administrator ↔ Test and appointment management
- Email Service ↔ Confirmation and notification messages

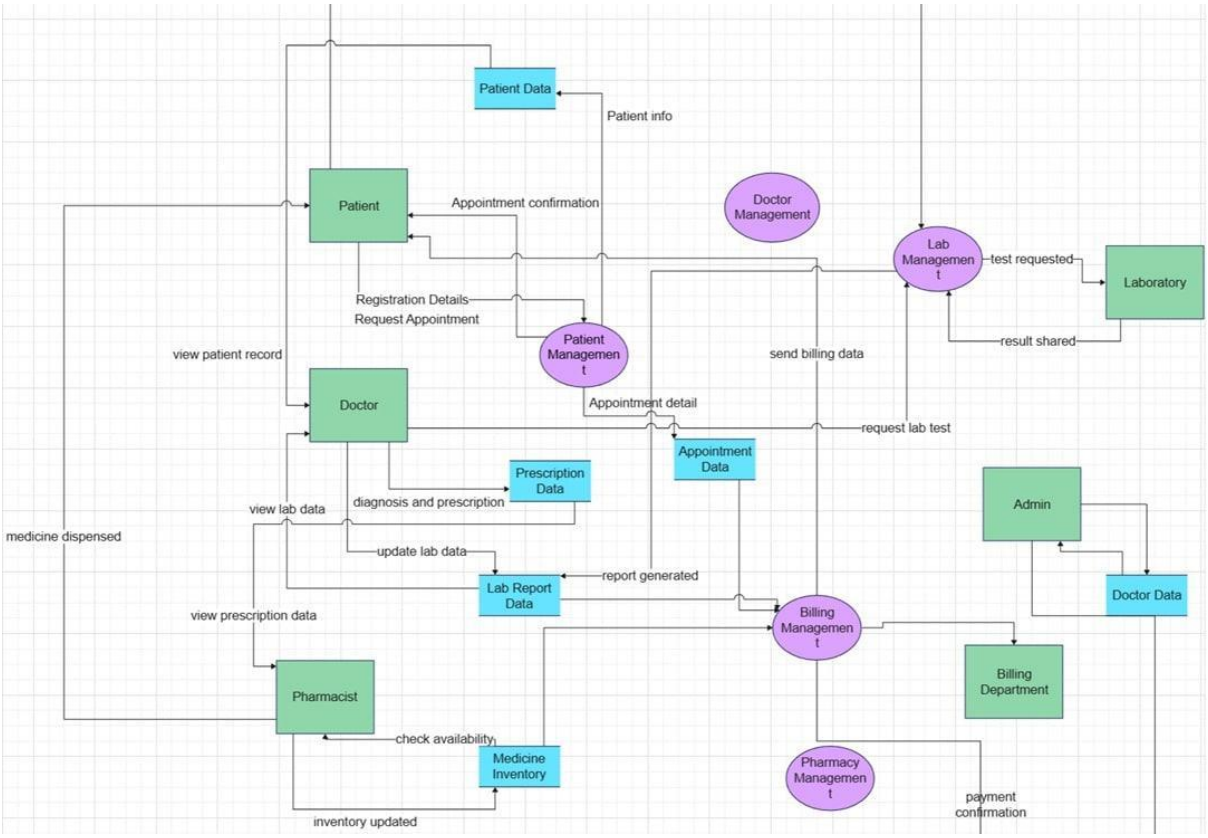
Level 1 – DFD Processes

1. **Manage Patient & Authentication**
2. **Manage Tests & Appointments**

3. **Process Payment**
4. **Generate Laboratory Report**
5. **View Reports through Service Portal**

Data Stores:

- D1 – Patient Data
- D2 – Test & Appointment Data
- D3 – Payment Data
- D4 – Report Data



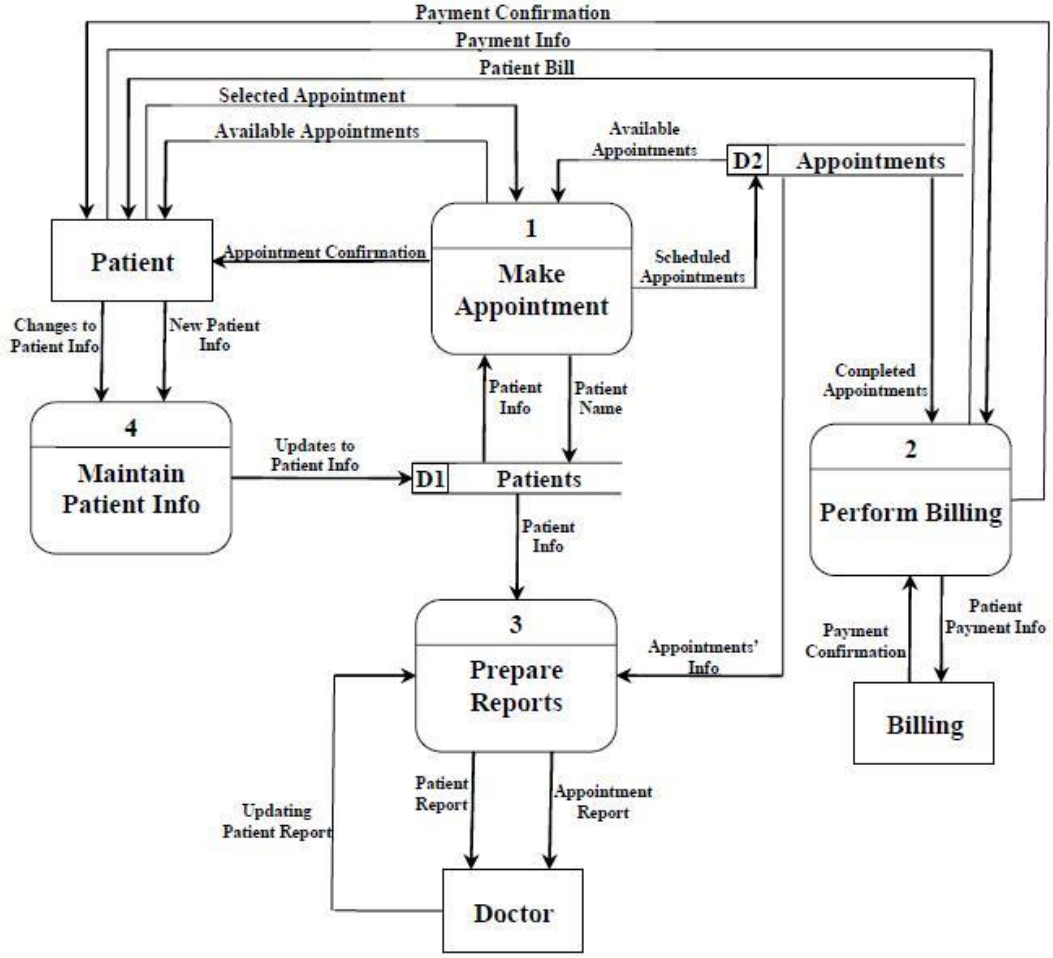
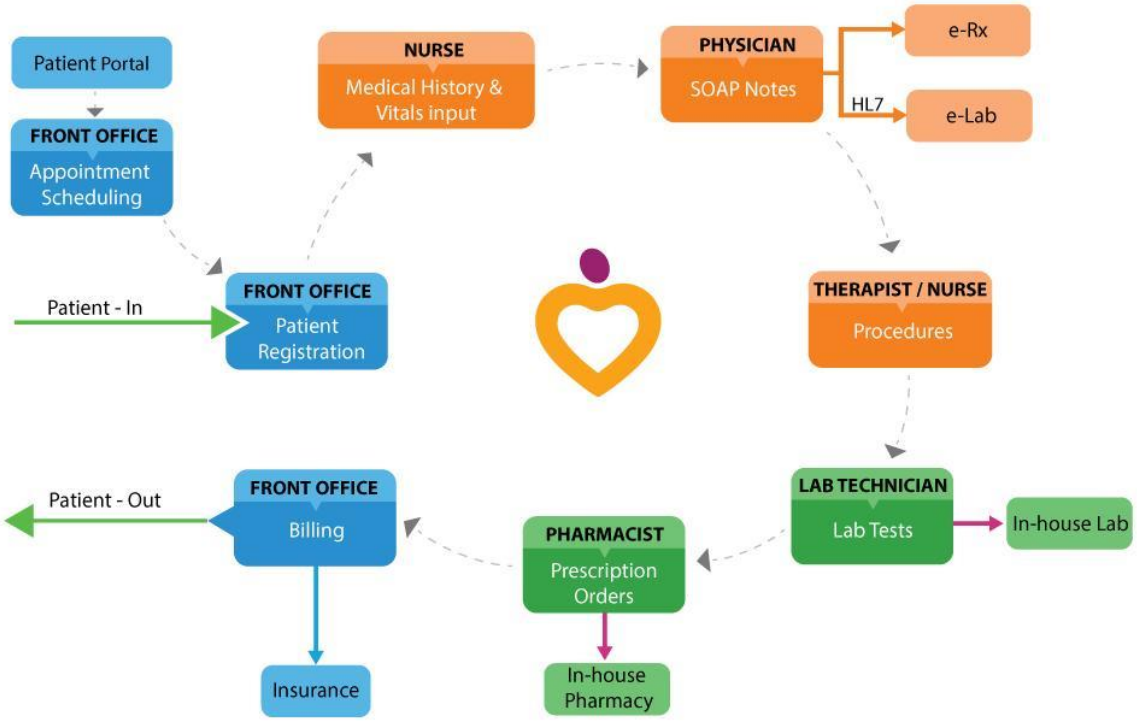


Fig. 2. Level-0 DFD for the Patient sub-system



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Patient	Registration / Login	USN-1	As a patient, I can access the Medical Diagnostic Center portal using my account.	I can successfully access the patient portal.	High	Sprint-1
Patient	Browse Tests	USN-2	As a patient, I can view the available diagnostic tests.	I can view the test name, category, description and price.	High	Sprint-1
Patient	Test Selection	USN-3	As a patient, I can select a diagnostic	The selected test is displayed	High	Sprint-1

			test that I want to book.	correctly before booking.		
Patient	Appointment Booking	USN-4	As a patient, I can book a diagnostic test by selecting an appointment date and time slot.	The appointment is created with the selected test, date and time.	High	Sprint-1
Patient	Patient Details	USN-5	As a patient, I can enter my personal details while booking a test.	Patient details are saved successfully with the appointment.	High	Sprint-1
Patient	Payment	USN-6	As a patient, I can complete payment for my selected diagnostic test.	Payment is processed and the payment status is updated successfully.	High	Sprint-2
Patient	Booking Confirmation	USN-7	As a patient, I can receive confirmation after successfully booking a test.	Confirmation contains the appointment details and test information.	High	Sprint-2
Patient	My Appointments	USN-8	As a patient, I can view my booked appointments and their status.	I can view pending, confirmed, completed or cancelled appointments.	High	Sprint-2

Patient	Appointment Cancellation	USN-9	As a patient, I can cancel an eligible appointment.	The appointment status changes to cancelled successfully.	Medium	Sprint-2
Patient	Lab Reports	USN-10	As a patient, I can view my laboratory reports after completing the diagnostic test.	The patient's available reports are displayed in the My Lab Reports page.	High	Sprint-3
Patient	Report Details	USN-11	As a patient, I can view detailed information about my laboratory report.	Report number, test name, date, doctor and findings are displayed.	High	Sprint-3
Doctor / Lab Technician	Test Results	USN-12	As a doctor or lab technician, I can enter and update diagnostic test results.	Test results are saved against the correct patient and appointment.	High	Sprint-3
Doctor / Lab Technician	Report Verification	USN-13	As a doctor, I can verify laboratory reports.	The report shows the verification details after verification.	High	Sprint-3
Administrator	Test Management	USN-14	As an administrator, I can add, update and	Diagnostic test information can be	High	Sprint-1

			manage diagnostic tests.	created and maintained successfully.		
Administrator	Appointment Management	USN-15	As an administrator, I can review and update patient appointment status.	Appointment status is updated correctly in the system.	High	Sprint-2
Administrator	Report Management	USN-16	As an administrator, I can manage patient laboratory reports.	Reports can be created, updated and associated with the correct patient.	High	Sprint-3
System	Notifications	USN-17	As a system, I can send notifications for important appointment events.	The patient receives the appropriate appointment notification.	Medium	Sprint-3
System	Automatic Report Generation	USN-18	As a system, I can automatically generate a laboratory report when an appointment is completed.	A report is automatically created and associated with the completed appointment.	High	Sprint-3

Overall User Journey

Browse Tests → Select Test → Enter Patient Details → Book Appointment → Make Payment → Confirmation → Test Completed → Laboratory Report Generated → View Report